

Sam Duong

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M.S. Computer Science student at Georgia Tech, where my research is on retrieval and verification for AI systems: when a grounded model cites the wrong source, and what it takes to catch it. I also cofounded a GeoAI company and build the product, so the same questions get tested in production. First author paper accepted at UrbanAI '26 (ACM SIGSPATIAL).

EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science

Atlanta, GA

Expected 2027

Master of Science in Urban Analytics, Concentration in Machine Learning and Spatial Analysis

2026

Georgia State University

Atlanta, GA

Bachelor of Science in Computer Information Systems, Concentration in Big Data Analytics

Jul 2024

EXPERIENCE

CTO and Founding AI Engineer | Jan 2025 - Present

Polymetron, Atlanta, GA

- Built and shipped the product with one cofounder: a **multi agent system of 5 agents and 16 tools** over **12,550** indexed building code passages, in TypeScript, Next.js and Python, deployed on Vercel.
- Kept the system honest under change by writing a **232 check regression suite** that replays finished agent runs with no model in the loop, so every release is gated in CI rather than spot checked by hand.
- Eliminated **all 74 wrong source citations** on held out cases while keeping 28 of 28 valid ones, by tracing each failure to retrieval and shipping a governance gate. Written up in a paper accepted at UrbanAI '26.
- Shipped a **12,933 parcel** public dashboard and a resident submission portal backed by **1,161 tests**, delivered with a neighborhood coalition and a nonprofit who used it in live public meetings.

Graduate Research Assistant and Geospatial Engineer | Aug 2025 - Present

Georgia Tech, Center for Urban Resilience and Analytics and I2CE Lab, Atlanta, GA

- Built a hybrid graph plus vector retrieval engine on Neo4j and pgvector and benchmarked it against vector only search on a **56 question ground truth set**, scoring accuracy, faithfulness and citation recall.
- Built the corpus and quasi experimental design for a difference in differences study of how building code amendments affect housing supply, structuring ordinances from **120+ U.S. cities** and matching treated to comparable control cities.
- Shipped a **30 layer** spatial platform and four production dashboards used at a workshop with **20+ organizations**, merging Census, USDA, CDC and FEMA data in PostGIS.

Data Analyst Associate | Jun 2024 - Jul 2025

Atlanta Beltline Inc., Atlanta, GA

- Shipped a vulnerability analysis covering all **45 neighborhoods** of the 22 mile corridor as a public dashboard, built in Python and R with the planning team.

Data Analytics and Engineering Intern | Aug 2023 - Apr 2024

Waystar Health, Atlanta, GA

- Cut recurring data extraction time **20%** in Snowflake SQL and helped the revenue team cut next quarter churn by **\$240,000** with five Power BI dashboards.

RESEARCH AND PUBLICATIONS

- **AI for Urban Buildability: An Atlanta Zoning Case Study on Where Citation-Bound Verification Fails.** Duong, S. M. and Rocker, I. M. Full paper, accepted at UrbanAI '26 (ACM SIGSPATIAL workshop); artifact public.
- **A High-Fidelity Urban Digital Twin for Resilient Urban Regeneration in Nihonbashi, Tokyo.** Full chapter, under review, CUPUM 2027 volume *Future Cities in the Era of AI*.
- **Who Can Get a Retrofit? Retrofit Access Across Georgia's 159 Counties.** In preparation for ACM COMPASS 2027.

SELECTED PROJECTS

Civic Evidence Platform · 360CITI Cambridge | 2026

- Designed and shipped a public platform for a Massachusetts city covering **12,933 parcels and 261 zones**, with a resident forum and submission portal (Next.js, TypeScript, Supabase, MapLibre, PMTiles). Draco compression took the tile payload from 7.4 MB to **1.6 MB**, which is what made it usable on phones.

Research Workspace Platform | 2026

- Built a multi tenant workspace for a research lab, backed by **672 tests** and a probe gated blue green deploy so a bad release never reaches users.

Open Data Analyses · 3 public repos | 2026

- A PySpark insurance GLM against an XGBoost challenger, a grid headroom what if on EIA 930, and a triple difference study of NYC congestion pricing over **344M trips** where placebo tests killed four candidate findings before write up.

LEADERSHIP AND AWARDS

- **First Place, TAG Data Challenge 2025** (led a team building a multi agent RAG chatbot for a nonprofit) | **Microsoft and Claude Campus Ambassador**, Georgia Tech (Fall 2026) | **Volunteer**, CodePath

TECHNICAL SKILLS

- **Building:** TypeScript, Python, SQL, R, React, Next.js, Node, FastAPI, PostgreSQL, PostGIS, Supabase, Neo4j, Vercel, Git, CI/CD, Playwright, Vitest, pytest
- **AI and research:** multi agent systems and tool use, RAG and vector retrieval, eval harness design, LLM as judge, causal inference, XGBoost, SHAP, PyTorch, Pandas, GeoPandas, PySpark, deck.gl, MapLibre